Independently Verifiable Stablecoin Liquidity Investigation Agent

Evidence-grounded investigation and independent replay of Ethereum Curve 3pool

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Abstract

This study tests whether evidence guidance and independent recomputation improve stablecoin investigation reports. We froze 51 block-pinned snapshots covering two Ethereum Curve 3pool windows and defined six tasks with 18 required components. Sixteen were answerable. We compared a deterministic template with ordinary and evidence-guided agents using the same DeepSeek Flash model, task data, tool access and budget limits.

Across 18 scheduled runs per model condition, the ordinary agent achieved reproducible correct coverage of 34/48 (70.8%) and the evidence-guided agent achieved 43/48 (89.6%). Verification withheld six nonconforming structured numeric submissions, one with an incorrect value and five with incorrect query bounds or fields. Published correct coverage was 31/48 and 42/48, respectively. On a separate 32-case test, the second-source verifier rejected 10 of 12 injected faults and returned UNKNOWN on the other two.

A retrospective replay found that verification intercepted one wrong research action and deferred eight correct ones. Small-order extrapolation overestimated the size-matched quote at 50 historical blocks. Quote screening decisions agreed at the five tested thresholds. [16]

Research questions

Generation. Does evidence guidance change raw correctness, reproducibility or model usage under matched settings?

Verification. Which claims survive a fixed verification and withdrawal rule, and how much correct coverage is lost?

Source retrieval. What does a second archive interface add over independent arithmetic using the original snapshot?

Investigation workflow

01 / Acquire	02 / Generate	03 / Recompute	04 / Publish
Resolve blocks at UTC targets and store raw responses.	Emit required answer slots, citations and a tool trace.	Check original-source and second-source evidence separately.	Retain PASS claims and record each withdrawal and reason.

The evidence package also contains a separate local-chain digest registration demonstration. [1, 2, 3]

Data and frozen benchmark

The Ethereum Curve 3pool holds DAI, USDC and USDT. Nominal balances use the tokens' ERC-20 balanceOf(pool) values. Each target maps to a block whose timestamp is at or before the target and whose next block is later. The logs preserve target and observed times, block identities, request parameters and raw responses. The primary archive used Alchemy and the comparison archive used dRPC. The source check establishes agreement across those interfaces for the captured observations. Their upstream infrastructure is unverified. [2]

Window and UTC sampling	Snapshots	Tasks
USDC depeg context, 10 Mar 2023 00:00 to 11 Mar 00:00 inclusive, plus 11 Mar 01:00 for the dependent quote	26	R01-R03
FTX context, 9 Nov 2022 00:00 to 10 Nov 00:00 inclusive. The dependent quote falls inside this grid.	25	R04-R06

Deterministic metrics

Normalize each integer balance by dividing by 10 raised to its decimals. A token's share is its fraction of the three normalized balances. The share change is the final percentage minus the initial percentage. The quote rate is normalized output divided by input. Size sensitivity is the signed difference between the 1,000-input rate and the 100,000-input rate, divided by the 1,000-input rate and multiplied by 10,000. Imbalance is half the sum of the absolute differences between each share fraction and one third. [2]

Task family	Required components	Boundary
R01 / R04	DAI share, 1,000 USDC-to-USDT quote rate, and relative size sensitivity	Fixed hour, direction and units
R02 / R05	USDC share change, sampled maximum imbalance, and quote one hour after the returned maximum	Maximum over allowed grid points
R03 / R06	First-hour USDC share, largest allowed one-hour increase, and whether continuous peak is identifiable	Increases checked from 18:00 to 00:00 UTC, skipping the excluded 22:00 point

Controls and denominator

T is a single deterministic report per task. G and E each ran six tasks three times in a recorded randomized order. E adds requirements for precise citations, checks on units and times, and limits on interpretation. The comparison evaluates this prompt as a whole. The 16 answerable slots contribute 48 opportunities to each model arm, including failed runs. Each complete task set also has two non-answerable slots. [2, 3]

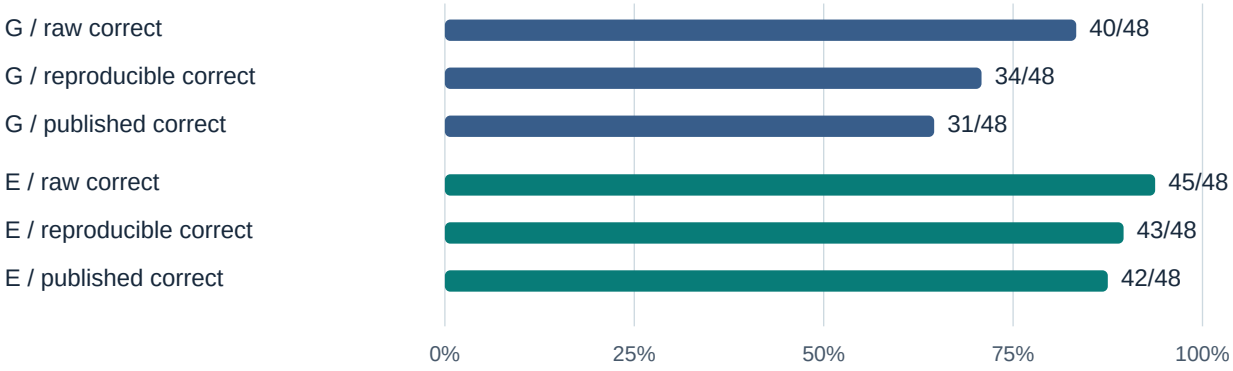
The ten exploratory rounds found a contrast in citation replayability and no clear difference in unsupported statements on the local pilot model. Those observations informed the choice of reproducible correct coverage before the formal G/E freeze. The metric rewards correct answers with complete attribution, which E explicitly requests. We also report raw correctness and the descriptive prose audit. [3, 8, 10]

The freeze used a team-attested consolidated reference workbook with two calculation paths. It matched 18 components and 51 snapshots. Reviewer names, dates and signatures are absent from that workbook. The attestation records that independent dual review was unverified. Page 5 describes the post-freeze checks. [2, 4, 12]

02 / FORMAL EXPERIMENT

Correct coverage and cost

Correct coverage requires the correct value, unit, time and query for the frozen slot. Reproducible correct coverage also requires a V1 PASS, including the exact evidence ID set, formula version and dataset digest. Publication uses the original model draws and the same denominator. [3, 6]



Measure	T template	G ordinary	E evidence
Scheduled runs / reports submitted	6 / 6	18 / 16	18 / 18
Correct coverage, raw	16/16 (100.0%)	40/48 (83.3%)	45/48 (93.8%)
Reproducible correct coverage, raw	16/16 (100.0%)	34/48 (70.8%)	43/48 (89.6%)
Correct coverage after publication	N/A	31/48 (64.6%)	42/48 (87.5%)
Withdrawn numeric items: value / query errors	0 / 0	1 / 2	0 / 3
Correct raw answers lost on publication	N/A	9/40 (22.5%)	3/45 (6.7%)
Over-refusals on answerable slots	0/16	0/48	0/48
Refusals passing the exact-reason check	2/2	0/6	0/6
Model calls / logged tokens	0 / 0	63 / 104,166	70 / 129,007

G's two stops omitted five answerable slots and one non-answerable slot. Across all 18 pairs, the reproducible gap was 18.75 percentage points. In the 16 pairs where both arms completed, raw correctness tied at 40/43, while reproducible correctness was 34/43 for G and 39/43 for E, a gap of 11.63 points. The completed-pair result describes that subset. [3, 14]

The frozen refusal scorer required the literal reason code. The prompt asked for a specific reason in words. By meaning, G gave five valid refusals across six scheduled non-answerable slots, with one omitted at a budget stop. E gave six valid refusals across six slots. [3, 8]

Matched limits and actual use

Client admission used the same ceilings of 48,000 input, 12,000 output and 60,000 total tokens, with 10 calls and 2,048 output tokens per call. All 133 provider receipts were within the reserved bounds. Archived generation wall times totaled 191.602 s for G and 234.587 s for E. [3, 11]

Five offline batches recorded medians of 0.216 s for V1, 0.283 s for V2 and 0.002 s for in-memory publication. A separate replay measured a 0.901 s median from saved input loading through checked publication. Prose review and adjudication took seven person-hours. Live elapsed time, billing and total project labor remain unreported. [11, 13, 15]

Where the checks succeed and fail

V1 recomputes claims using the frozen primary archive. V2 checks the allowed observations and block identities against a second RPC capture and applies the same verifier arithmetic. Both use arithmetic separate from the agent's calculation and check units, scope, evidence and numerical tolerance before publication. [5, 6]

Case set	Cases	V1 PASS / FAIL / UNKNOWN	V2 PASS / FAIL / UNKNOWN
Injected errors	12	5 / 7 / 0	0 / 10 / 2
Clean twins	12	12 / 0 / 0	12 / 0 / 0
Legitimate refusal scenarios	4	0 / 0 / 4	0 / 0 / 4
Missing secondary source	2	2 / 0 / 0	0 / 0 / 2
Unresolved source conflict	2	2 / 0 / 0	0 / 0 / 2

V2 rejected 10/12 injected faults. The three additional detections involved block identity. The other two injected faults returned UNKNOWN because source evidence was missing. Both layers accepted all 12 clean twins. The design schedules 136 case or record checks across V1 and V2, including the two empty model records in both layers. This count is separate from claim verdicts and RPC traffic. [5]

Verification and withdrawal

The captures agree on the bound fields, and V2 changed none of the 102 formal claim verdicts. Totals were 34 PASS, 3 FAIL and 11 UNKNOWN for G and 43 PASS, 2 FAIL and 9 UNKNOWN for E. Of the 12 correct claims withheld, eight had incomplete attribution and four depended on a withdrawn claim. The publisher constructs a new answer from surviving fields. The experiment measures structured withdrawals. [3, 5, 6]

Correcting the slot-to-claim mapping restored three G and four E items. The corrected replay records UNKNOWN for source-conflict cases U3 and U4 and supersedes the original fault summary for those cases. The formal outcomes are unchanged by that correction. The archive retains the original reports and verdict records. [3, 5, 6]

Evidence integrity on a local chain

The local Ganache demonstration registers the evidence manifest digest and reads it back. It rejects duplicate registration and produces a different, unregistered digest for a modified manifest. The saved receipt has chain ID 1337 and block number 2. Checking covered artifacts requires comparing their bytes with the manifest hashes. The arithmetic is checked by the separate verifier. [7]

Independent arithmetic before publication

Across both windows, V1 and V2 accepted six controls and rejected six reports with injected denominator, token-decimal or extrema errors. With the same error injected into both verifiers, all six matching wrong reports originally passed and were published. The added exact-fraction check withheld all six and two quotes tied to erroneous peak times, while retaining all six controls. The check covers balance shares, changes and sampled imbalance maxima. Quote arithmetic continues under V2. [17]

04 / INTERPRETATION

What the evidence supports

Prose evidence audit

Of the 42 scheduled records, 34 contained model prose. Six template stubs and two G budget stops sit outside the prose denominators. Two sets of annotations cover all 34 reports. He Yuequn confirmed the reconciliation workbook on 10 November 2026. Its fixed-component judgments match the frozen-data review. [8, 13]

Prose evidence measure	G ordinary	E evidence
Numerical bundles matching frozen data	42/43	48/48
Numerical bundles with incomplete original citations	6/43	2/48
Supported scope-limited refusals observed	5/5	6/6

The two reviews contain 123 and 120 annotations. Claim counts differ in 19/34 reports. The team holds 62 reconciliation and explanation rows and a 143-unit follow-up packet. Common-list agreement and unsupported-claim rates are pending. A separate rule-based initial audit found at least one unsupported assertion in 11/16 G reports and 6/18 E reports. [8, 13, 15]

A worked example

For R02, the allowed endpoint snapshots put USDC at 29.679648% and 48.100138% of the pool's nominal balances. The difference is +18.420491 percentage points. The sampled maximum imbalance is 0.277696115 at 11 March 2023 00:00 UTC. The 100,000 USDC-to-USDT quote rate at the required following hour is 0.9897830738 USDT/USDC. The frozen packet identifies endpoint blocks 16794061 and 16801143. [2, 8]

P41 reports a decrease of about 18.42 percentage points in both the structured item and the prose. Both verifiers reject the sign error, and the item is withheld. The endpoint balances show a share increase. Attributing that change to capital flows, solvency or a specific mechanism requires additional evidence. [2, 3, 8]

Reference checks received after the freeze

WANG Hanyang (A) matched 18/18 components. WU Yuxuan (B) matched 17/18 after converting three proportions to percentages. The combined workbook includes A's formulas and saved results. The timing of B's access to A's work remains unverified. [12]

For R06/S2, B recorded 1.11692608956 percentage points. He Yuequn adjudicated the discrepancy on 9 November 2026 and accepted A's value of 1.16926089569553 percentage points, which matches the frozen reference used for formal scoring. The workbook preserves the original entries. The separate review record documents the decision. Review isolation remains unverified. [12]

Scope of the findings

Scheduled reproducible coverage was 15/24 versus 22/24 in the USDC window and 19/24 versus 21/24 in the FTX window, for G and E respectively. The gap was larger in the USDC window under both counting rules. The study covers one pool and six tasks with repeated runs. Its second-source results apply to the recorded interfaces and scripted faults. [2, 3, 5, 14]

05 / PRACTICAL EVALUATION

Historical decision replay

Research decisions from the same reports

We applied three rules to archived reports before and after verification. The rules labelled net USDC share direction, selected the sampled peak hour and escalated eligible hourly share increases of at least 1 percentage point. Recomputing from the two matching frozen datasets yielded the same benchmark actions. Missing or withdrawn inputs went to review. All 36 scheduled decisions include the one affected by a budget stop. [16]

Research action	Raw correct / wrong / review	Verified correct / wrong / review
Label direction	11 / 1 / 0	11 / 0 / 1
Select follow-up hour	12 / 0 / 0	7 / 0 / 5
Flag concentration rise	11 / 0 / 1	8 / 0 / 4
All scheduled decisions	34 / 1 / 1	26 / 0 / 10

Verification caught the wrong decreasing label in R02 and sent nine additional decisions to review. Eight of the referrals involved correct actions with incomplete citations, including five peak-hour choices. Review time and eventual resolutions were outside the scope of the replay. [16]

Screening a 100,000 USDC-to-USDT quote

At 50 historical hourly blocks we screened a hypothetical order using either 100 times the 1,000 USDC quote or the archived quote for the full order size. The rule accepted quoted output at or above the stated minimum. The two source datasets agree at each block. [16]

Haircut, bps	Min. output, USDT	Proxy / size-matched eligible	Decisions changed
1	99,990	40 / 40	0/50
5	99,950	47 / 47	0/50
10	99,900	47 / 47	0/50
25	99,750	48 / 48	0/50
50	99,500	48 / 48	0/50

The proxy overestimated output at all 50 blocks. The median gap was 0.029438 USDT and the maximum was 2.872343 USDT at block 16,801,143. For that block, the output floor would have to be above 99,202.468457 and at or below 99,205.340800 USDT to change the action. The five illustrated floors produced the same decisions. [16]

The study covers one pool, one trade direction and two historical stress windows. The quote screen compares block-pinned quotes and excludes fills, gas, transaction ordering and returns. The research actions are rule-based replays of repeated model outputs. [16]

Replay and supporting records

Replay from the submission root

Action	Command
Replay deterministic data and verification	`python3 -B docs/reproduce_current.py`
Check saved formal V1, V2 and publication outcomes	`python3 -B -m curve_audit replay`
Replay the 32 frozen fault cases	`python3 -B docs/replay_formal_faults.py .`
Replay research and quote-screening decisions	`python3 -B audit/decision_replay_20261112/replay.py .`
Check packaged file hashes	`python3 -B docs/verify_delivery.py .`
Run the Python test suite	`python3 -B -m unittest discover -s tests -v`

Python 3.10 or later supports offline replay of the saved evidence. The packaged code matches 25 of 31 freeze-time module hashes. Six revisions and one added arithmetic module are documented in docs/code_provenance.json. Formal replay checks archived inputs and outputs. The archive also contains the freeze-time sources. [2, 3, 5, 15]

Development optimization after the formal run

The compact read-window response preserves raw balances, quotes and block identifiers. Three recorded budget stops passed a counterfactual pre-dispatch admission check with the smaller payload. In two nine-run development samples using the same tasks and budget, completed submissions rose from 8/9 to 9/9 and measured token use fell from 66,475 to 62,078. The samples used different model draws and the improvement is reported separately from the formal comparison. [9]

Evidence index

[1] Course Trade 2 scope and instructor proposal feedback, 14 October 2026.
[2] results/current_preparation/{binding, dataset, secondary_dataset, tasks}.json and data/current_study/ source records.
[3] experiments/formal_v1/{freeze, result_summary, publication_replay}.json and saved raw report traces.
[4] results/current_preparation/reference_review/review_attestation.json and consolidated workbook.
[5] experiments/formal_v1/verifier_replay_20261105.json, corrected 32-case fault replay.
[6] curve_audit/current_verify.py and curve_audit/verification.py.
[7] contracts/anchor.mjs and results/current_chain/anchor.json.
[8] experiments/formal_v1/prose_review_input.json and audit/{prose_audit_summary, prose_audit_rows}.json.
[9] results/optimization_20261105/{summary, budget_replay}.json.
[10] references/trial-report-20261102.md, exploratory trial and metric selection rationale.
[11] audit/RESOURCE_USE.md and audit/resource_use_20261109.json, recorded usage and packaged replay timings.
[12] audit/reference_review_20261109/REVIEW.md, post-freeze reference review and evidence limits.
[13] audit/prose_review_20261110/REVIEW.md, human review and reconciliation provenance.
[14] audit/FORMAL_SENSITIVITY.md and audit/compare_run_sets.py, scheduled and completed-pair results.
[15] README.md, LIMITATIONS.md, REPORT_ERRATA.md and docs/code_provenance.json.
[16] audit/decision_replay_20261112/, historical decision replay.
[17] audit/arithmetic_faults/README.md, commands and results for both arithmetic fault replays.

Freeze SHA-256 `ec66a6aaf06a0a37eeff5e74bbda6e68a866be00617a00a9413087679e7cd220`